

Information Sheet

Instructor

Prof. Hye Won Chung

Office: N1-206

E-mail: hwchung@kaist.ac.kr

Course E-mail: ee326.26spring@kaist.ac.kr

Teaching Assistants

Yongmin Lee

Hongsin Lee

Junmo Kim

Junhyung Kim

Hyunmin Cho

Lectures:	Wednesday & Friday 9:00-10:15am, N1-201
Prof.'s Office Hours:	Wednesday 4-5pm at N1-206
TA's Office Hours:	Monday 4-5pm at N1-203

Welcome to EE326 !

This course offers a undergraduate-level introduction to information theory. Information theory is probably one of the most elegant mathematical theories, with the most direct and significant engineering impacts to our lives in the information age. Starting from the first paper by Claude E. Shannon ("Mathematical theory of communications," Shannon, 1948), information theory has found its applications in many areas, including communications, machine learning, statistics, and data science, etc. Among all those applications, the focus of this course will be on the direct applications of information theory to digital communications and machine learning.

Many information theorist, including myself, believe that the most important aspect of learning information theory is to learn a new way of thinking in modeling and solving engineering problems. In this sense, this course would be beneficial not only to students who have interests in communications or machine learning but also to students in other engineering disciplines. As one of a few mathematical theory-based undergraduate classes in EE, we will proceed with a combination of intuitive thinking and somewhat rigorous mathematical treatments.

Prerequisites

Basic knowledge of probability, at the level of EE210, is necessary. The effective prerequisite is *fluency* with basic quantitative probabilistic reasoning and analysis. A good book to review probability is

D. Bertsekas and J. Tsitsiklis, *Introduction to Probability*.

Textbook and Lecture Notes

We will not follow one unique book, but the basic contents of this course and some homework problems will be taken from

T. M. Cover and J. A. Thomas. *Elements of Information Theory*, Wiley, 2nd ed., 2006.

Instead of following one unique book, I will upload lecture notes every one or two weeks. One lecture note would cover 2-4 lectures.

Classes and Office Hours

In this course, we plan to meet in person at N1-201 on 9:00am-10:15am every Wednesday and Friday. But for some weeks (less than 10 %) when I am out of town, the lectures will be possibly replaced by online lecture videos uploaded at KLMS.

I will offer an office hour every Wednesday 4-5pm at N1-206. Any students who have questions on the course materials are welcome to join.

Course Webpage and Problem Sets

Webpage: <http://klms.kaist.ac.kr/>

Every student who registered for this class can obtain course materials from KLMS. Please check KLMS often enough to see whether there is a new lecture note.

Problem sets (PSETs) will be given roughly every week except the first week and the midterm week. Usually, PSETs will be **issued on Wednesday (sometimes Friday)** and will be **due on the next week by 4pm**. Please submit your PSETs through KLMS webpage, so that TAs can grade your PSETs. PSET solutions will be available at KLMS just after the due, so only the PSETs submitted in time will be accepted. Every PSET will include 2-3 problems.

Exams and Course Grades

There will be two exams in this course. Dates for the exams are **Monday, Apr. 20, 9am–11:30am, and Monday, June 15, 9am–11:30am**. I will update you the exam details as the exam dates are approaching.

The final grade in the course will be based upon our best assessment of your understanding of the material during the semester. Roughly, the weights used in grade assignment will be:

Midterm	40%
Final	40%
Homework	10%
Attendance	10%

Syllabus

Tentatively, the contents of the course would include the following:

- Definitions and properties of information theoretic quantities, including entropy, mutual information and divergence
- Lossless source coding: Huffman codes, Shannon codes
- Method of types, strong typicality, large deviation
- Joint AEP and channel coding theorem
- Source channel separation theorem and feedback channel
- Applications of information theory to other disciplines: data science/machine learning

Syllabus and Schedule

	Date	Topic	PS due	PS out
W	3/4	L1: Preliminaries and introduction		
F	3/6	L2: Probability review		
W	3/11	L3: Entropy		1
F	3/13	L4: Relative entropy and mutual information		
W	3/18	L5: Convexity, concavity of information measures	1	2
F	3/20	L6: Fano's inequality, data processing inequality		
W	3/25	L7: Huffman codes	2	3
F	3/27	L8: Asymptotic equipartition property (AEP)		
W	4/1	L9: Lossless source coding I	3	4
F	4/3	L10: Lossless source coding II		
W	4/8	L11: Method of types	4	5
F	4/10	L12: Strong typicality		
W	4/15	L13: Universal source coding		
F	4/17	L14: Review	5	
M	4/20	Midterm (through L14 and PS5)		
W	4/29	L15: Capacity of discrete memoryless channel (DMC)		6
F	5/1	L16: Joint AEP & channel coding theorem		
W	5/6	L17: Channel coding theorem (converse)	6	7
F	5/8	L18: Channel coding theorem (achievability)		
W	5/13	L19: Source channel separation & Feedback channels	7	8
F	5/15	L20: Rate distortion theory I		
W	5/20	L21: Rate distortion theory II	8	9
F	5/22	L22: Info. theory in machine learning I		
W	5/27	L23: Info. theory in machine learning II	9	10
F	5/29	No class		
W	6/3	Holiday		
F	6/5	L24: Info. theory in machine learning III		
W	6/10	L25: Info. theory in machine learning IV		
F	6/12	L26: Review	10	
M	6/15	Final (through L26 and PS10)		